

PPO–Based Trajectory Optimization for Rechargeable Agricultural Robots in Greenhouse Micro–Climatic Sensing

Introduction

Greenhouses exhibit complex micro-climatic variations due to differences in plant canopy, ventilation, sunlight exposure, and structure. High-resolution climate monitoring is essential for precision agriculture but difficult to achieve using static sensors due to cost, maintenance issues, battery limitations, and humidity damage.

This work presents a **complete PPO-based trajectory optimization framework** that enables a rechargeable agricultural robot to collect climate measurements efficiently, safely, and autonomously in realistic greenhouse settings.

Research Gap

Traditional wireless sensor networks are unsuitable for long-term greenhouse monitoring. Earlier PPO-based approaches showed promise but lacked realistic battery modeling, detailed greenhouse graph structures, dynamic reward design, robustness testing, and reproducible implementation.

Our work overcomes these limitations by delivering a complete, practical, and deployable RL system for energy-aware agricultural sensing robots

Methodology (PPO)

The system uses **Proximal Policy Optimization (PPO)** with:

- Actor–Critic neural network
- Categorical policy over next-node movements
- Clipped surrogate loss for stable updates
- Generalized Advantage Estimation (GAE)
- Entropy regularization for exploration
- Gradient clipping & minibatch updates

Real greenhouse climate datasets are used to generate node-specific information variability, guiding the robot’s sensing priorities.

Reward Function & Training Logic

Reward integrates:

- **+ Information Gain** → prioritize high-variance Pols
- **– Movement Energy Cost**
- **– Revisits Penalty**
- **+ Charging Bonus**
- **Critical Battery Penalty (exponential)** → prevents unsafe depletion

This ensures:

- energy-aware navigation
- diverse sampling
- safe and reliable recharging
- optimized climate data collection

Results & Evaluation

Climate Prediction Accuracy

Using PPO-generated trajectories improves temperature prediction accuracy by **2–2.5%** compared to baseline routes.

Energy Sustainability

Across 1656 trials:

- With **300 battery units** → 257 charging stops
- With **150 battery units** → 406 stops **(+58%)**
Agent adapts automatically to reduced battery capacity.

Safety Performance

Sharifi et al. baseline:

- **83 unsafe battery depletion events**
Our PPO system:
- **0 depletion events**

Robustness

When greenhouse paths were modified, PPO recalculated efficient routes autonomously, showing real-world adaptability.

Improvements

This work introduces several advancements over prior PPO-based greenhouse navigation:

- **Dynamic, state-aware reward shaping**
- **Adaptive energy thresholds**
- **Expanded state space** (including time, history, and energy usage patterns)
- **Enhanced action space** (partial charging, speed adaptation)
- **Curriculum learning + dynamic hyperparameters**
- **Improved climate prediction accuracy: 3.5–4%**
- **Scalable to larger layouts and multi-robot scenarios**
- **Modular, documented, fully reproducible code**

Conclusion

This research delivers the **first fully implemented, practically deployable PPO-based trajectory optimization system** for rechargeable greenhouse sensing robots. By integrating:

- realistic battery physics,
- a structured graph-based environment,
- a mathematically grounded reward function, and
- stable PPO training procedures,

the system significantly improves climate sensing quality, energy sustainability, and operational safety.

The PPO agent consistently identifies high-value regions, avoids battery depletion, adapts to environmental changes, and supports downstream climate prediction models with higher-quality data.

The study establishes deep reinforcement learning as a strong, scalable approach for long-term agricultural monitoring in controlled environments.

Future extensions include multi-agent coordination, wireless charging integration, real-time re-planning, and physical robot deployment.